

HW8 Solutions

ECE 310: Digital Signals Processing, Spring 2025

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Solution:

- (a) For the sine part, $\omega_1 = \frac{\pi}{5}$, $\Omega_1 = 5\pi$, $T = \frac{\omega_1}{\Omega_1} = \frac{1}{25}\text{s}^{-1}$. Similarly, for the cosine part, $\omega_2 = \frac{2\pi}{5}$, $\Omega_2 = 10\pi$, $T = \frac{\omega_2}{\Omega_2} = \frac{1}{25}\text{s}^{-1}$. Therefore, a choice for T is $\frac{1}{25}\text{s}^{-1}$.
- (b) No. For the sine part, there are multiple possible digital frequencies: $\omega_1 = \frac{\pi}{5} + 2k\pi$. Similarly, for the cosine part, $\omega_2 = \frac{2\pi}{5} + 2\ell\pi$. In order to share the same sampling period for both sine and cosine part, i.e., $T = \frac{\omega_1}{\Omega_1} = \frac{\omega_2}{\Omega_2}$, we need

$$\frac{\omega_1}{\omega_2} = \frac{\Omega_2}{\Omega_1} = \frac{1}{2}.$$

Therefore,

$$\frac{\omega_1}{\omega_2} = \frac{\frac{\pi}{5} + 2k\pi}{\frac{2\pi}{5} + 2\ell\pi} = \frac{1}{2}$$

leads to $2k = \ell$. One possible sampling rates are $\omega_1 = \frac{11\pi}{5}$ and $\omega_2 = \frac{22\pi}{5}$, which means

$$T = \frac{\omega_1}{\Omega_1} = \frac{11}{25}\text{s}.$$

Grading: 12 points

- +5 for the correct answer in (a)
- +3 for stating non-unique in (b)
- +4 for the correct answer in (b)

2

Solution:

For an ideal D/A, $g_a(t - nT) = \text{sinc}(\frac{\pi(t-nT)}{T})$ (Here it's the **unnormalized** sinc function)

$$\begin{aligned} x_c &= x[-1] \frac{\sin(\frac{\pi(t+T)}{T})}{\frac{\pi(t+T)}{T}} + x[3] \frac{\sin(\frac{\pi(t-3T)}{T})}{\frac{\pi(t-3T)}{T}} \\ &= -2 \frac{\sin(\frac{\pi(t+T)}{T})}{\frac{\pi(t+T)}{T}} + 5 \frac{\sin(\frac{\pi(t-3T)}{T})}{\frac{\pi(t-3T)}{T}} \end{aligned}$$

Grading: 10 points

- +6 correct ideal D/A formula.
- +4 correct output.

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Solution:

a)

$$f_{Nyquist} = 2B = 2 * 5\text{kHz} = 10\text{kHz}$$

$$T = \frac{1}{f_{Nyquist}} = .1\text{ms}$$

b) To make sure the system behaves as an LSI system, we only need to ensure no aliasing from $\omega \in [-\frac{\pi}{8}, \frac{\pi}{8}]$.

To do this, we set up an equation where the left side is the maximum positive component of the signal in the digital spectrum and the right side is maximum value that won't alias with the second spectral copy:

$$2\pi BT = 2\pi - \frac{\pi}{8}$$

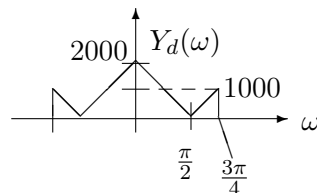
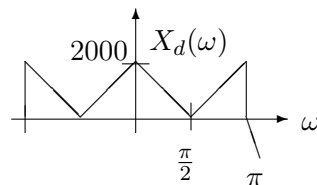
$$B = \frac{1}{2\pi T} (2\pi - \frac{\pi}{8})$$

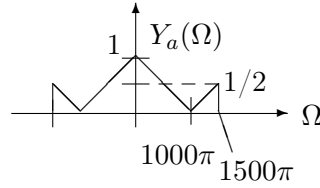
$$B = \frac{1}{T} - \frac{1}{16T} = 10000 - \frac{10000}{16} = 9.375\text{kHz}$$

Grading: 15 points

- +5 Max T value
- +5 Recognizing some aliasing is allowed
- +5 Correct maximum bandwidth

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Solution:



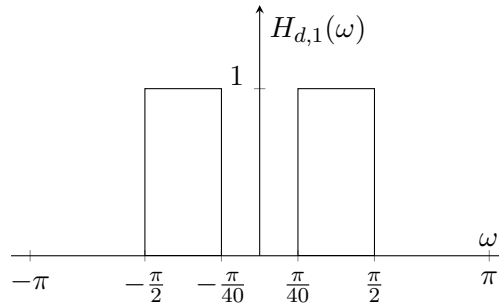
Grading: 18 points

- +6 each plot

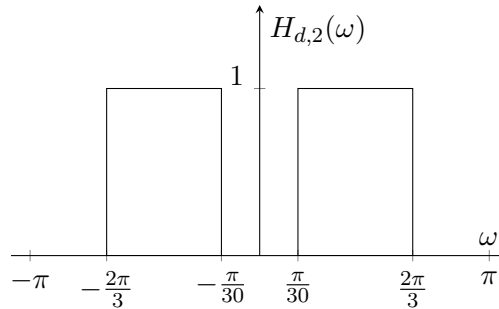
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Solution:

- a) $F_{nyquist} = 2F_{x_a} = 24kHz$
- b) 300 Hz in CTFT will be mapped to $\frac{2\pi \cdot 0.3}{24} = \frac{\pi}{40} rad/sample$ in DTFT. 6 kHz in CTFT will be mapped to $\frac{2\pi \cdot 6}{24} = \frac{\pi}{2} rad/sample$ in DTFT. The frequency response of $H_{d,1}(\omega)$ can be plotted as below.



- c) To perform the desired filtering function, we should prevent any aliasing under 6 kHz. Suppose the sampling period is T_s . 12kHz in CTFT will be mapped to $24\pi T_s$ in DTFT. 6kHz in CTFT will be mapped to $12\pi T_s$ in DTFT. $-24\pi T_s + 2\pi \geq 12\pi T_s$, which will result as $T_s \leq \frac{1}{18} msec$.
- d) 300 Hz in CTFT will be mapped to $\frac{2\pi \cdot 0.3}{18} = \frac{\pi}{30} rad/sample$ in DTFT. 6 kHz in CTFT will be mapped to $\frac{2\pi \cdot 6}{18} = \frac{2\pi}{3} rad/sample$ in DTFT. The frequency response of $H_{d,2}(\omega)$ can be plotted as below.



Grading: 21 points

- +3 Correct Nyquist rate

- +6 Correct $H_{d,1}$
- +6 Correct largest T
- +6 Correct $H_{d,2}$

6

Solution:

(a) **Correct.** By the definition of DFT:

$$\begin{aligned}
 X[19] &= \sum_{n=0}^{20} x[n] e^{-j \frac{2\pi k n}{21}} \Big|_{k=19} = \sum_{n=0}^{20} x[n] e^{-j \frac{38\pi n}{21}} \\
 &\stackrel{(a)}{=} \sum_{n=0}^{20} x[n] e^{-j \frac{38\pi n}{21} + j 2\pi n} = \sum_{n=0}^{20} x[n] e^{j \frac{4\pi n}{21}} \\
 &\stackrel{(b)}{=} X_d \left(-\frac{4\pi}{21} \right)
 \end{aligned}$$

where (a) comes from the 2π periodicity and (b) holds by the definition of DTFT.

(b) **Correct.** Again, by the definition of DFT:

$$\begin{aligned}
 X[2] &= \sum_{n=0}^{20} x[n] e^{-j \frac{2\pi k n}{21}} \Big|_{k=2} = \sum_{n=0}^{20} x[n] e^{-j \frac{4\pi n}{21}} \\
 &\stackrel{(a)}{=} \left(\sum_{n=0}^{20} x^*[n] e^{j \frac{4\pi n}{21}} \right)^* \stackrel{(b)}{=} \left(\sum_{n=0}^{20} x[n] e^{j \frac{4\pi n}{21}} \right)^* \\
 &= X_d^* \left(-\frac{4\pi}{21} \right)
 \end{aligned}$$

where (a) comes from the property of complex conjugate, (b) comes from the fact that $x[n]$ is real-valued therefore $x[n] = x^*[n], \forall n$.

(c) **Incorrect.** Again, by using the same definition as in part (a):

$$X[12] = \sum_{n=0}^{20} x[n] e^{-j \frac{2\pi k n}{21}} \Big|_{k=12} = \sum_{n=0}^{20} x[n] e^{-j \frac{24\pi n}{21}} = X_d \left(\frac{24\pi}{21} \right)$$

Then there is no clue that $X_d \left(\frac{24\pi}{21} \right) = X_d \left(-\frac{4\pi}{21} \right)$. Indeed, suppose the sequence is $x[n] = \delta[n-1]$, then one can see that $X_d \left(\frac{24\pi}{21} \right) = e^{-j \frac{24\pi}{21}}$ and $X_d \left(-\frac{4\pi}{21} \right) = e^{j \frac{4\pi}{21}}$, which is a counter-example that the two values are not equal.

(d) **Incorrect.** Again, by using the same definition as in part (b):

$$X[4] = \sum_{n=0}^{20} x[n] e^{-j \frac{2\pi k n}{21}} \Big|_{k=4} = \sum_{n=0}^{20} x[n] e^{-j \frac{8\pi n}{21}} = \left(\sum_{n=0}^{20} x[n] e^{j \frac{8\pi n}{21}} \right)^* = X_d^* \left(-\frac{8\pi}{21} \right)$$

There is no clue that $X_d^* \left(-\frac{8\pi}{21} \right) = X_d^* \left(-\frac{4\pi}{21} \right)$. Indeed, if we still plug in $x[n] = \delta[n-1]$, then we have $X_d^* \left(-\frac{8\pi}{21} \right) = e^{-j\frac{8\pi}{21}}$ and $X_d^* \left(-\frac{4\pi}{21} \right) = e^{-j\frac{4\pi}{21}}$, which are not equal.

Grading: 12 points

- +1 each correct answer and +2 each reasonable justification

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Solution:

By observation, one can convert x to y by first scaling x by $1/2$, which yields an intermediate sequence z such that

$$z = \frac{1}{2}x = [1, 0, 3, 2]$$

Next, we (circularly) shift z to the right by 1 to obtain y . That is,

$$y[n] = z[\langle n-1 \rangle_4]$$

In terms of the relation on DFT, first by linearity we have the relation between $Z[k]$ and $X[k]$ that

$$Z[k] = \frac{1}{2}X[k]$$

where $Z[k]$ is the DFT of z at the k -th frequency. Then, by the (circular) time shift property of DFT:

$$Y[k] = Z[k]e^{-j\frac{2\pi k}{4}} = Z[k]e^{-j\frac{\pi k}{4}} = \frac{1}{2}e^{-j\frac{\pi k}{4}}$$

In this way, we can express the DFT of y as $Y = [Y_0, Y_1, Y_2, Y_3]$ such that

$$Y_0 = \frac{1}{2}X_0, \quad Y_1 = -j\frac{1}{2}X_1, \quad Y_2 = -\frac{1}{2}X_2, \quad Y_3 = j\frac{1}{2}X_3,$$

Grading: 12 points

- +2 identified shift
- +2 identified scale
- +4 correctly applied properties
- +4 correct result